MATHEMATICAL WHITE PAPER

Failure of Wang–Landau convergence under unrestricted monotone refinement

An explicit eleven-state counterexample with a quantitative probability bound

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Abstract. Monotone decrease of the Wang–Landau modification factor to one does not, by itself, ensure consistency of the density-of-states estimator. We construct an eleven-state system with two energy levels, uniform symmetric proposals, the usual acceptance and update rules, and an 80% histogram-flatness threshold. For the constant-root schedule $f_k = \exp(10^{-7k})$, with flatness checked after each update, an event of probability greater than 0.24 has the following properties: every stage terminates with a flat histogram, the modification factor tends to one, and the estimated density ratio converges to an incorrect limit with relative error greater than 15%. The proof controls all future adaptation by a summable displacement budget and bounds stage durations using a martingale argument at deterministic auxiliary horizons. A related construction also fails with checks every 10,000 updates. This is a concrete counterexample to an unrestricted schedule-sufficiency interpretation of the original algorithmic claim; error saturation itself is established prior art.

Keywords: Wang–Landau algorithm; density of states; adaptive Monte Carlo; histogram flatness; consistency; martingale concentration.

1 The claim and its scope

Wang and Landau describe an adaptive Monte Carlo algorithm that estimates the number of microstates at each energy. In Section II of their 2001 paper, after discussing convergence and square-root reduction of the modification factor, they state:

> *"In fact, any function may be used as long as it decreases f monotonically to 1."*

The adjacent schedule is $f_{k+1} = f_k^{1/n}$ for $n > 1$. The same discussion recognizes that accuracy also depends on the flatness criterion and implementation details. The assertion is informal, rather than a formal theorem with a specified probability guarantee. We examine its unrestricted interpretation: that monotone refinement, together with the stated adaptive rules and a fixed percentage-flatness test, suffices for consistency. [1, Section II, printed pp. 3–4]

The counterexample uses an intentionally aggressive member of that constant-root family. Its asymptotic conclusion concerns indefinitely continued execution in exact real arithmetic. It does not evaluate the empirical performance reported for the usual square-root choice. Belardinelli and Pereyra previously analyzed error saturation under exponential refinement and proposed power-law refinement. The present finite construction makes no claim to discovery of that literature defect. [2]

2 The finite system and algorithm

Let the microstate space be $\Omega = \{0, \ldots, 10\}$. Assign energy 0 to states 0 through 4 and energy 1 to states 5 through 10. The true density of states is therefore $g_*(0) = 5$ and $g_*(1) = 6$. The parameter to estimate is the normalization-independent ratio

$$R_* = g_*(1)/g_*(0) = 6/5,$$
$$D_* = \log R_* = \log(6/5).$$

This can be viewed as a single classical spin with eleven labels and a two-valued energy function. Start at any fixed microstate. At each update, independently propose a uniformly selected microstate. Thus the proposal kernel is $q(x, y) = 1/11$, including self-proposals. It is symmetric and connects the entire space.

Maintain log estimates $S(0)$ and $S(1)$, initially both zero, so that $\hat{g}(i) = \exp(S(i))$. At stage k, beginning with $k = 0$, use

$$r = 10^{-7}, \quad a_k = r^k, \quad f_k = \exp(a_k).$$

Consequently $f_0 = e$, $f_{k+1} = f_k^{1+1/10{,}000{,}000}$, and f_k decreases strictly to one.

1. Given current state X and proposal Y, accept the proposal with probability $\min\{1, \exp[S(E(X)) - S(E(Y))]\}$. Use independent acceptance uniforms.
2. Let Z be the retained state, including a rejected move. Add a_k to $S(E(Z))$ and add one to the corresponding stage histogram count.
3. After each update, let $s = H(0) + H(1)$. End the stage at its first positive time satisfying

$$\min\{H(0), H(1)\} \geq (4/5)(s/2) = 2s/5. \quad (1)$$

At termination, reset both histogram counts to zero and increment the stage index. The log estimates are retained. Rule (1) selects 80% within the paper's percentage-flatness definition. Both accepted and rejected moves update the estimator. A stage that never meets (1) continues at its current increment indefinitely. These rules define a process at every finite update time.

Write S_t for the estimates after t total updates, and define

$$D_t = S_t(1) - S_t(0), \quad \hat{R}_t = \exp(D_t).$$

An additive normalization of the two log estimates leaves both quantities unchanged.

3 Main result

Theorem 1. For the process in Section 2, there is an event $\mathscr{B}$ with probability

$$\mathrm{P}(\mathscr{B}) \geq (30/121)[1 - 2e^{-20}/(1 - e^{-10})] > 0.24, \quad (2)$$

on which every stage terminates, infinitely many stages are completed, and D_t converges to a finite limit. For every $t \geq 2$ on this same event,

$$\begin{gathered} |D_t| < 1/100, \\ |D_t - \log(6/5)| > 17/100, \\ |\hat{R}_t - 6/5|/(6/5) > 47/297 > 15\%. \quad (3) \end{gathered}$$

In particular, the estimator does not converge in probability to the correct ratio, and its probability of convergence to that ratio is strictly less than one.

The event is shared across all future times. The result therefore establishes persistent error, rather than a collection of isolated finite-time deviations.

4 Proof of the main result

4.1 A positive-probability initial state

Let A be the event that the first proposal has energy 0 and the second has energy 1. The first proposal is accepted because the estimates are equal. After updating energy 0, the log estimates are (1, 0); the second proposal is accepted with probability one. Hence

$$\mathrm{P}(A) = (5/11)(6/11) = 30/121.$$

The first histogram (1, 0) fails (1), while (1, 1) passes. On A, stage 0 therefore lasts exactly two updates and leaves $S_2(0) = S_2(1) = 1$, so $D_2 = 0$.

4.2 A deterministic budget for future adaptation

For stages $k \geq 1$, set $T_k = 8000(k + 1)$. Let τ_k be the duration of stage k; set $\tau_k = \infty$ if that stage is never reached or never terminates. Each update changes D by either $+a_k$ or $-a_k$. If every stage respects its duration budget, the total absolute change after time 2 is at most

$$\begin{aligned} B &= \textstyle\sum_{k\geq 1} T_k a_k \\ &= 8000r(2 - r)/(1 - r)^2 \\ &= 159999992000/99999980000001 < 1/100. \quad (4) \end{aligned}$$

This bounds intermediate updates as well as stage endpoints. It also bounds the variation accumulated through any fixed stage and a hypothetical continuation of that stage to its deadline.

4.3 Conditional transition probabilities

Suppose $|D| \le 1/100$ before an update. Every cross-energy acceptance probability is at least $\exp(-1/100)$. Put $q_0 = 5/11$ and $q_1 = 6/11$. Conditional on the complete past, the probability p_i of the next retained energy being i satisfies

$$q_i e^{-1/100} \le p_i \le 1 - q_{1-i} e^{-1/100}.$$

If the current energy differs from i, entry is a proposal to i followed by acceptance. If the current energy is i, retention is one minus the exit probability. The displayed bounds cover both cases. Since $e^{-1/100} \ge 99/100$, both energies obey

$$9/20 \le p_i \le 11/20. \quad (5)$$

These are conditional bounds for an adaptive process. They require neither independent retained energies nor a binomial model for counts observed at a stopping time.

4.4 Bounding a stage duration without assuming it terminates

Condition on A and a full history at the start of stage k for which all earlier deadlines succeeded. Define an auxiliary process that uses the same proposals and acceptance uniforms but holds the increment at a_k for exactly T_k updates, ignoring any earlier flatness trigger.

The deterministic bound

$$|D| \le \sum_{1 \le j < k} T_j a_j + T_k a_k \le B < 1/100$$

holds along every auxiliary path through that horizon. Consequently (5) holds before every auxiliary update. No conditioning on future deadline success has been used.

Let I_ℓ indicate that auxiliary update ℓ retains energy 1, and let p_ℓ be its conditional expectation given the auxiliary past. Then

$$M_s = \sum_{\ell=1}^{s} (I_\ell - p_\ell)$$

is a martingale with increments bounded in absolute value by one. Its predictable count at the deadline lies between $9T_k/20$ and $11T_k/20$. A nonflat terminal histogram has its energy-1 count below $2T_k/5$ or above $3T_k/5$. Either case forces $|M_{T_k}| \ge T_k/20$. The bounded-increment martingale inequality proved in Appendix A gives

$$\begin{aligned} &\mathrm{P}(\text{auxiliary histogram nonflat at } T_k \mid \text{starting history}) \\ &\quad \le 2\exp(-T_k/800) = 2\exp[-10(k+1)]. \quad (6) \end{aligned}$$

If the actual stage lasts longer than T_k, it has not yet triggered flatness. Its history then agrees with the auxiliary process throughout the horizon, and the auxiliary histogram must be nonflat at that horizon. Thus (6) also bounds the conditional probability of $\tau_k > T_k$.

4.5 Infinitely many successful deadlines

Let G_m mean that $\tau_k \le T_k$ for every $1 \le k \le m$. Decompose failure according to the first missed deadline and apply (6) conditional on the corresponding starting history:

$$P(G_m \mid A) \ge 1 - \sum_{k=1}^{m} 2 \exp[-10(k+1)].$$

The events G_m decrease with m. Continuity of probability and the geometric series therefore give

$$P(\cap_{m\ge1} G_m \mid A) \ge 1 - 2e^{-20}/(1 - e^{-10}).$$

Taking $\mathscr{B} = A \cap \cap_{m\ge1} G_m$ proves (2). The bound is strictly greater than 0.24 using only rational arithmetic: $e > 2$ implies that the geometric tail is less than $2/(1024 \times 1023)$, and

$$(30/121)[1 - 2/(1024 \times 1023)] = 2618875/10562816 > 6/25.$$

On $\mathscr{B}$, every stage has finite duration. Each requires at least two updates, so infinitely many stages cannot accumulate at a finite update time. Equation (4) shows that the total variation of D_t after time 2 is finite. Hence D_t converges to a finite limit and stays within 1/100 of zero.

4.6 Persistent estimation error

For $x > 0$, integrating $1/(1 + u) > 1 - u$ gives $\log(1 + x) > x - x^2/2$. Thus

$$\log(6/5) > 1/5 - (1/5)^2/2 = 9/50.$$

On $\mathscr{B}$, the log-ratio error consequently exceeds $9/50 - 1/100 = 17/100$ for every $t \ge 2$. Moreover, $\exp(x) \le 1/(1 - x)$ for $0 \le x < 1$, by comparison of their power series. Therefore

$$\hat{R}_t < \exp(1/100) \le 100/99 < 6/5,$$
$$|\hat{R}_t - 6/5|/(6/5) > 1 - (5/6)(100/99) = 47/297 > 3/20.$$

This completes the proof of (3). For every deterministic $t \ge 2$, the probability of log-ratio error exceeding 0.17 is greater than 0.24. Hence convergence in probability fails; almost-sure convergence to the correct ratio fails as well. In fact, on $\mathscr{B}$ the limit exists and is incorrect. □

5 Fixed intervals between flatness checks

The paper reports checking histograms approximately every 10,000 Monte Carlo sweeps in its experiments. The main construction chooses checks after every update; the next variant shows that frequent checking is not essential to the consistency obstruction.

Fix an even positive integer L, and check flatness only at update counts $L, 2L, \ldots$ within each stage. In this single-spin system an update is a sweep. Replace the increment ratio and deadlines by

$$r_L = 1/(10^7 L), \quad a_k = r_L{}^k,$$
$$T_k = 8000L(k+1).$$

Let A_L be the event that the first L proposed energies alternate 0, 1, 0, 1, …. Both moves of every pair are accepted. At the start of each pair the estimates are equal; after its energy-0 move, the next energy-1 move has acceptance probability one. Thus $P(A_L) = (30/121)^{L/2}$, and stage 0 ends at its first check with equal estimates.

Writing $r = 10^{-7}$, the subsequent displacement budget satisfies

$$B_L = 8000Lr_L(2 - r_L)/(1 - r_L)^2$$
$$\le 16000r/(1 - r)^2 < 1/100.$$

Every deadline is an allowed check time, so the same coupling and martingale proof applies. There is a persistent-error event of probability at least

$$(30/121)^{L/2}[1 - 2e^{-20L}/(1 - e^{-10L})] > 0. \quad (7)$$

In particular, consistency fails for the variant with $L = 10{,}000$. Its displayed lower bound is extremely small. The probability greater than 24% in Theorem 1 applies specifically to checking after each update, not to this fixed-interval variant.

6 Interpretation and finite stopping

The failure combines a permissive flatness threshold with a schedule that removes almost all future capacity to adapt. When the estimated density ratio is one, the true energy frequencies are approximately 0.455 and 0.545. Both already lie inside the interval [0.4, 0.6] accepted by (1). Repeated histogram tests can therefore succeed while the density ratio is materially wrong. The proof quantifies the possibility that this continues forever.

For the illustrative stopping threshold $\log(f_{\text{final}}) = 10^{-8}$, the main schedule has $a_1 = 10^{-7}$ and $a_2 = 10^{-14}$. Stopping when the factor first falls below that threshold, or after completing that next stage, still produces more than 15% ratio error on $\mathscr{B}$. The asymptotic proof couples these finite outputs to the indefinite process. A tiny terminal modification factor is therefore insufficient by itself to certify a tiny estimation error.

The result isolates a missing condition in the unrestricted interpretation of the schedule assertion. It does not establish failure of every Wang–Landau schedule, a $1/t$ refinement, or the specific experimental measurements in the original paper. Nor does it provide a complete characterization of schedules that converge.

7 Reproducibility and workspace grounding

The accompanying standard-library Python checker, `verify_wl.py`, verifies 25 exact arithmetic statements supporting the proof. Its receipt, `wl_verification.json`, records the constants, check results, and SHA-256 hashes of the derivation and relevant source files. It uses rational arithmetic without Monte Carlo simulation or floating-point approximations.

Verification component	What it establishes
Displacement budget	The exact fraction in (4), finite-series and tail decompositions, and the uniform budget for fixed check intervals.
Transition and concentration constants	The rational conditional-probability bounds, flatness margins, and the exponent coefficient 1/800.
Probability and error bounds	The rational lower bound above 24%, the log-error threshold, and the relative-error bound 47/297.
Scope control	The same budget does not certify the ordinary square-root schedule; no conclusion about that schedule follows from this construction.

Run the checker from the accompanying directory:

```
py -3.12 -B verify_wl.py
```

The Python checker validates arithmetic certificates. The accompanying Lean development now verifies the main stochastic counterexample. `WangLandau.closed_counterexample` has no initialization, stage-tail, or drift assumptions: all three are proved for the explicit measurable adaptive sampler. `closed_bad_limit_event` also verifies infinite stage progress, the executed factor limit 1, and convergence to a wrong estimate on a measurable event of probability greater than 24%. The fixed-check-interval and finite-stopping extensions remain outside the Lean formalization.

The formal proof uses an equivalent bound through independent reference counts. Mark a proposal of energy i when its acceptance draw is below 0.99. The two reference probabilities are 0.45 and 0.54. While $|D| < 0.01$ every marked proposal is accepted, so the actual histogram dominates both reference counts. Independent Hoeffding bounds give probability at most $2\exp(-T/800)$ that either count falls below $0.4T$. A simultaneous induction maintains the drift budget and count domination: a good row of draws forces flatness by its deadline, and the histogram reset reserves the next stage's budget. A countable first-failure bound then gives the same probability and error constants as the martingale proof.

The formal sampler allocates independent noise by stage and within-stage update. A causal masking proof verifies that each adaptively consumed draw has the original uniform proposal and acceptance law, independently of the full history of consumed draws. Thus row allocation does not introduce a different sampling rule.

Replay the Lean proof and axiom audit with:

```
powershell -NoProfile -ExecutionPolicy Bypass -File .\replay_lean.ps1
```

The build receipt `lean_verification.json` records compiler exits, source hashes, theorem names, and scope.

`lean_build.log` records axiom dependencies. Only the standard foundations `propext`, `Classical.choice`, and `Quot.sound` are permitted; there are no admitted proofs or added axioms.

The construction develops three existing workspace distinctions. The flat-histogram tolerance work separates a bound conditional on a good bias estimate from a theorem that a sampler learns that estimate. The detailed-balance work distinguishes inversion of collected statistics from stochastic convergence. The concentration work rejects treating adaptively selected visit counts as independent binomial observations. These ideas motivate the displacement budget and conditional martingale argument here. The self-contained proof does not take earlier workspace verdict labels as mathematical assumptions.

Relevant antecedents: `human_formula_flat_histogram_tolerance_receipt.md`; `human_formula_wl_tmmc_db_receipt.md`; and `human_formula_tmmc_fixed_kernel_log_ratio_concentration_receipt.md`. The original detailed derivation is retained as `wl_derivation.txt`.

Appendix A The concentration inequality used in the proof

Let X_j be martingale differences with $|X_j| \le 1$ and conditional mean zero. For $\lambda \ge 0$ and $-1 \le x \le 1$, convexity gives

$$\exp(\lambda x) \le [(1+x)/2]e^{\lambda} + [(1-x)/2]e^{-\lambda}.$$

Taking conditional expectations yields a bound of $\cosh(\lambda)$. The inequality $\cosh(\lambda) \le \exp(\lambda^2/2)$ follows by integrating $(\log\cosh\lambda)' = \tanh\lambda \le \lambda$ for $\lambda \ge 0$. Iterated conditional expectation therefore gives

$$\mathrm{E}\,\exp(\lambda \textstyle\sum_{j=1}^{T} X_j) \le \exp(T\lambda^2/2).$$

Markov's inequality and the choice $\lambda = u/T$ imply

$$\mathrm{P}(\textstyle\sum_{j=1}^{T} X_j \ge u) \le \exp[-u^2/(2T)].$$

Apply the same argument to $-X_j$ and add the two bounds. With $u = T/20$, the result is $2\exp(-T/800)$, as used in (6). Every horizon in this application is deterministic after conditioning on the stage's starting history.

References

[1] F. Wang and D. P. Landau. "Determining the density of states for classical statistical models: A random walk algorithm to produce a flat histogram." *Physical Review E* **64**, 056101 (2001). doi:10.1103/PhysRevE.64.056101. Author preprint: arXiv:cond-mat/0107006. The quoted schedule assertion appears on printed page 4.

[2] R. E. Belardinelli and V. D. Pereyra. "Wang–Landau algorithm: A theoretical analysis of the saturation of the error." *Journal of Chemical Physics* **127**, 184105 (2007). doi:10.1063/1.2803061. Author preprint: arXiv:cond-mat/0702414.